

Figure S1. Donor-balanced bootstrap analysis of EC subtypes showing one of the iterations as an example. (A) Number of EC sampled during balanced downsampling ($n = 50$ cells per donor) grouped per donor (left) and age group (right). (B) Number of RAB13+ EC per individual after downsampling. (C) UMAP embedding of downsampled EC colored by donor identity. (D) UMAP representation of downsampled EC split into young and aged group, colored by subtype. (E) Stacked bar plots representing the proportion of each EC subtype per individual (left panel) and age group (right panel). (F) Relative differences in cell proportions for each vascular state, comparing young and aged EC. Red and blue represent clusters statistically significant ($FDR < 0.05$ and absolute $\log_2$ fold change > 1) in young and aged EC, respectively. (G,H) Stacked bar plots representing the proportion of cells in each EC subtype per individual (G) and age group (H). (I) UMAP plot illustrating the recomputed dimensional reduction and clustering of downsampled EC colored by subtype (left) and cluster (right). (J) Heatmap showing Jaccard similarity index between the recomputed clusters (X axis) and EC subtypes (Y axis). (K) Clustering robustness metrics across iterations, including Jaccard similarity (left) and Adjusted Rand Index (ARI) (K), demonstrating stability of the RAB13+ EC cluster.

Figure S2. Donor-balanced bootstrap analysis of MSC subtypes showing one of the iterations as an example. (A) Number of MSC sampled during balanced downsampling ($n = 50$ cells per donor) grouped per donor (left) and age group (right). (B) Number of THY1+ Fibro-MSC per individual after downsampling. (C) UMAP plot illustrating the distribution downsampled MSC colored by individual. (D) UMAP representation of downsampled MSC split into young and aged group, colored by subtype. (E) Stacked bar plots representing the proportion of each MSC subtype per individual (left panel) and age group (right panel). (F) Relative differences in cell proportions for each MSC population, comparing young and aged MSC. Red and blue represent clusters statistically significant ($FDR < 0.05$ and absolute $\log_2$ fold change > 1) in young and aged MSC, respectively. (G, H) Stacked bar plots representing the proportion of cells in each MSC subtype per individual (G) and age group (H). (I) UMAP plot illustrating the recomputed dimensional reduction and clustering of downsampled MSC colored by subtype (left) and cluster (right). (J) Heatmap showing Jaccard similarity index between the recomputed clusters (X axis) and MSC subtypes (Y axis). (K) Clustering robustness metrics across iterations, including Jaccard similarity (left) and Adjusted Rand Index (ARI) (K), demonstrating stability of the THY1+ fibro-MSC cluster.

Figure S3. Concordance between donor-level and single-cell differential expression analysis. (A) Correlation of $\log_2$ fold-change estimates for common RAB13+ EC marker genes between single-cell and pseudobulk analyses. (B) Correlation of $\log_2$ fold-change estimates for common THY1+ fibro-MSC marker genes between single-cell and pseudobulk analyses. (C) Correlation of $\log_2$ fold-change estimates for common age-associated DEGs between single-cell and pseudobulk analyses (Spearman $\rho = 0.89$). (D) Correlation of $\log_2$ fold-change estimates for common age-associated DEGs between single-cell and pseudobulk analyses (Spearman $\rho = 0.89$).

Figure S4. Sensitivity analysis of endothelial cell populations after exclusion of dominant donor Elderly 4. (A) UMAP projection of EC after exclusion of donor Elderly 4, colored by donor identity (left) and endothelial subtypes (right). (B) Differential abundance analysis showing fold changes in endothelial subtypes between aged and young donors following donor exclusion, with FDR-adjusted p-values indicated. (C) Stacked bar plots representing the proportion of each EC subtype per age group. (D) Stacked bar plots representing the proportion of cells in each EC subtype per individual (left) and age group (right). (E) Recomputed UMAP embeddings after donor exclusion colored by donor identity (left), endothelial subtype (middle), and cluster assignment (right). (F) Heatmap of cluster correspondence between the endothelial subtypes from the original dataset and the re-assigned clusters. (G) Venn diagram showing overlap of RAB13+ EC marker genes identified in the original dataset versus after exclusion of Elderly 4.

Figure S5. Sensitivity analysis of endothelial cell populations after exclusion of dominant donor Elderly 3. (A) UMAP projection of MSC after exclusion of donor Elderly 3, colored by donor identity

(left) and mesenchymal subtypes (right). **(B)** Differential abundance analysis showing fold changes in mesenchymal subtypes between aged and young donors following donor exclusion, with FDR-adjusted p-values indicated. **(C)** Stacked bar plots representing the proportion of each mesenchymal subtype per age group. **(D)** Stacked bar plots representing the proportion of cells in each mesenchymal subtype per individual (left) and age group (right). **(E)** Recomputed UMAP embeddings after donor exclusion colored by donor identity (left), mesenchymal subtype (middle), and cluster assignment (right). **(F)** Heatmap of cluster correspondence between the mesenchymal subtypes from the original dataset and the re-assigned clusters. **(G)** Venn diagram showing overlap of THY1+ fibro-MSC marker genes identified in the original dataset versus after exclusion of Elderly 3.

Figure S6. Analysis of mouse contribution to control, MGUS, and MM BI_{cy1} EC and MSC using sex-specific genes. **(A)** PCA plot showing separation of control EC based only on the expression of sex-specific genes colored by disease (upper panel) and Y-chromosome gene expression, where blue denotes cells with more than one count in Y-chromosome genes (bottom panel). **(B)** PCA plot showing separation of control EC based only on the expression of Y-chromosome genes colored by disease (upper panel) and Y-chromosome gene expression, where blue denotes cells with more than one count in Y-chromosome genes (bottom panel). **(C,D)** Similar to A,B for MGUS EC. **(E,F)** Similar to A,B for MM EC. MM EC cells were split into MM and MM2 clusters. **(G,H)** Similar to A,B for Control MSC. **(I,J)** Similar to A,B for MGUS MSC. **(K,L)** Similar to A,B for MM MSC. MM MSC cells were split into MM IFN low and high score. **(M)** Histogram of number of EC plotted by male to female gene ratios (scored continuously along the x-axis) in control, MGUS, and MM BI_{cy1} EC. MM EC cells were split into MM and MM2 clusters. Three discrete peaks emerge, representing cells likely originating from either sex or those with indeterminate ratios due to technical drop-off in scRNA-seq reads. **(N)** Similar to M for MSC.

